

Template System Implementation Checklist

Date: 2026-02-15

Source Guide: [docs/development/template-system-comprehensive-guide.md](#)

0) Preparation

- ☐ Confirm source documents are available and unchanged:
 - ☐ [docs/development/template-system-comprehensive-guide.md](#)
 - ☐ [docs/specs/template-system-requirements.md](#)
 - ☐ [docs/specs/built-in-template-expansion-requirements.md](#)
 - ☐ [docs/specs/built-in-template-automation-test-suite-requirements.md](#)
- ☐ Confirm implementation scope (template system only, no unrelated UI redesign).
- ☐ Confirm target environment assumptions (Windows + WSL2 local validation).

1) Metadata and Catalog Management

Files:

- [config/templates.json](#)
- [config/templates/*](#)

Tasks:

- ☐ Verify each built-in template has required base metadata:
 - ☐ [Id](#)
 - ☐ [Name](#)
 - ☐ [Category](#)
 - ☐ [Description](#)
 - ☐ [CompatibleDistros](#)
 - ☐ [Scripts](#)
- ☐ Verify extended metadata consistency where applicable:
 - ☐ [InstallMode](#)
 - ☐ [VersionOptions](#)
 - ☐ [DefaultSelections](#)
 - ☐ [PreflightChecks](#)
 - ☐ [OutputArtifacts](#)
 - ☐ [ScenarioTags](#)
- ☐ Ensure script assets exist for each [ScriptPath](#) referenced in metadata.
- ☐ Ensure script paths are relative and constrained under [config/templates/](#).

Definition of Done:

- ☐ Catalog metadata is complete, valid, and script paths resolve correctly.

2) Core Model and Service Contract

Files:

- `src/Client/DistroNexus.Core/Models/Template.cs`
- `src/Client/DistroNexus.Core/Interfaces/ITemplateService.cs`

Tasks:

- ☐ Confirm model coverage for metadata schema:
 - ☐ Version options model
 - ☐ Preflight checks model
 - ☐ Output artifact model
 - ☐ Install mode enum
- ☐ Confirm service interface includes:
 - ☐ Catalog load/search/get
 - ☐ Apply/validate/compatibility checks
 - ☐ Import/export custom templates
 - ☐ Application history retrieval

Definition of Done:

- ☐ Service contract fully represents template lifecycle operations.

3) Core Execution Implementation

Files:

- `src/Client/DistroNexus.Core/Services/TemplateService.cs`

Tasks:

- ☐ Confirm catalog loading strategy (AppData cache + local fallback).
- ☐ Confirm effective variable merge order:
 - ☐ `Variables`
 - ☐ `DefaultSelections`
 - ☐ `VersionOptions.DefaultValue`
 - ☐ Runtime overrides
- ☐ Confirm preflight check flow:
 - ☐ Required checks fail-fast
 - ☐ Optional checks warn and continue
- ☐ Confirm ordered script execution and timeout handling.
- ☐ Confirm `ContinueOnError` behavior works as intended.
- ☐ Confirm progress reporting via `TemplateProgress`.
- ☐ Confirm history persistence to `%APPDATA%\DistroNexus\template-application-history.json`.
- ☐ Confirm path safety protections:
 - ☐ Absolute path rejected
 - ☐ Path traversal blocked
 - ☐ Allowed root validation enforced

Definition of Done:

- ☐ Template apply pipeline is safe, observable, and deterministic.

4) Desktop Workflow Integration

Files:

- `src/Client/DistroNexus.Desktop/App.xaml.cs`
- `src/Client/DistroNexus.Desktop/Wizard/InstallWizardWorkflowViewModel.cs`
- `src/Client/DistroNexus.Desktop/Wizard/Steps/SelectTemplateStep.cs`
- `src/Client/DistroNexus.Desktop/Wizard/Steps/TemplateApplyStep.cs`

Tasks:

- ☐ Confirm `ITemplateService` DI registration exists.
- ☐ Confirm wizard includes template selection and apply steps in workflow.
- ☐ Confirm selected template and variables pass into apply phase correctly.
- ☐ Confirm template output/progress is visible in apply step UX.
- ☐ Confirm cancel/error paths behave gracefully.

Definition of Done:

- ☐ End users can select and apply templates inside install wizard reliably.

5) PowerShell Command Surface

Files:

- `src/PowerShell/DistroNexus.psd1`
- `src/PowerShell/Public/Get-DistroNexusTemplate.ps1`
- `src/PowerShell/Public/Apply-DistroNexusTemplate.ps1`
- `src/PowerShell/Public/Invoke-DistroNexusTemplateAutomation.ps1`

Tasks:

- ☐ Confirm cmdlet export in module manifest:
 - ☐ `Get-DistroNexusTemplate`
 - ☐ `Apply-DistroNexusTemplate`
 - ☐ `Invoke-DistroNexusTemplateAutomation`
- ☐ Confirm template discovery by ID/category works.
- ☐ Confirm template apply supports variable overrides.
- ☐ Confirm local automation runner supports:
 - ☐ `AllTemplates` mode
 - ☐ `SelectedTemplates` mode
 - ☐ `DryRun`
 - ☐ CI guard + override
 - ☐ Capability-gated classification (`Blocked`)

Definition of Done:

- ☐ PowerShell surface supports discovery, apply, and local automation validation.

6) Built-in Catalog Coverage Review

Files:

- `config/templates.json`

Tasks:

- ☐ Verify current built-in template set and category mapping:
 - ☐ `Development`
 - ☐ `Platform`
 - ☐ `CloudNative`
 - ☐ `Database`
 - ☐ `DataAndAI`
- ☐ Confirm all listed built-ins have metadata + scripts + basic probe readiness.
- ☐ Record gap status for `DevOps` category target (if still missing).

Definition of Done:

- ☐ Built-in coverage status is explicit and traceable.

7) Validation and Evidence

Tasks:

- ☐ Execute selective template automation run (single template).
- ☐ Execute selective template automation run (multiple templates).
- ☐ Execute full-catalog run when environment permits.
- ☐ Collect and verify artifacts under:
 - ☐ `docs/development/testing/results/<yyyymmdd>/<run-id>/summary.md`
 - ☐ XML result file (`NUnitXml` or `JUnitXml`)
 - ☐ Run manifest JSON
- ☐ Verify pass/fail/blocked classifications are accurate.

Definition of Done:

- ☐ Evidence package is complete and reproducible.

8) Documentation and Handoff

Tasks:

- ☐ Update template guide if behavior/schema changed.
- ☐ Update release notes / website docs if user-visible template capabilities changed.
- ☐ Summarize changed files and unresolved risks.

Definition of Done:

- ☐ Template-system implementation changes are fully documented and auditable.

Implementation Result

- ☐ Completed with no functional deviations.
- ☐ Completed with documented deviations.

- ☐ Blocked (requires follow-up).